

Leverage Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Leverage Expansion Intensity (LEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LEI achieves an annualized gross (net) Sharpe ratio of 0.52 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (20) bps/month with a t-statistic of 3.39 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets) is 19 bps/month with a t-statistic of 2.67.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices, yet the mechanisms through which production decisions affect expected returns remain central to understanding cross-sectional variation in risk premia. While extensive literature documents that firms' investment and financing choices predict future stock returns, the economic channels linking leverage adjustments to risk compensation in production-based economies require further investigation. We identify a critical gap in understanding how the intensity of leverage expansion maps to firms' marginal costs of production and their exposure to aggregate productivity shocks. This paper examines whether Leverage Expansion Intensity (LEI), measured as the rate of change in firms' debt financing relative to assets, captures systematic variation in expected returns through its connection to production frictions and time-varying risk premia in the cross-section of equities.

In production-based asset pricing models, firms' leverage decisions directly reflect the shadow value of capital and the marginal costs of adjusting productive capacity [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). When firms expand leverage intensively, they signal lower adjustment costs and higher marginal productivity of capital, which rational models predict should associate with lower expected returns due to reduced exposure to systematic productivity shocks [Liu et al. \(2009\)](#). The theoretical framework builds on the insight that leverage expansion intensity proxies for firms' investment opportunities and their ability to exploit positive productivity shocks without facing severe financing frictions [Golan et al. \(2013\)](#).

Firms experiencing rapid leverage expansion operate closer to their optimal capital structure, suggesting lower costs of external finance and reduced exposure to credit market disruptions [Hennessy and Whited \(2007\)](#). These firms exhibit greater operational flexibility, allowing them to smooth cash flows across states of nature and reducing their systematic risk exposure [Berk et al. \(1999\)](#). The production-

based perspective predicts that high LEI firms carry lower risk premia because their aggressive financing indicates efficient capital allocation and lower sensitivity to aggregate investment shocks.

The cross-sectional pricing implications emerge through the connection between leverage expansion and conditional betas with respect to production-based factors [Lin and Zhang \(2013\)](#). Firms with high leverage expansion intensity demonstrate lower loadings on investment and profitability factors, as their financing decisions reveal optimized production functions with diminished marginal returns to additional capital [Hou et al. \(2015\)](#). This mechanism generates predictable variation in expected returns, where intensive leverage expansion signals reduced compensation for bearing systematic production risk.

We find strong evidence that Leverage Expansion Intensity predicts cross-sectional stock returns with remarkable consistency across various specifications and robustness tests. The value-weighted long-short LEI strategy, which buys high LEI stocks and sells low LEI stocks, generates an average monthly return of 27 basis points with a t-statistic of 3.73, yielding an annualized Sharpe ratio of 0.52. The strategy’s alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t-statistic of 3.39, demonstrating that LEI captures unique variation in expected returns beyond established risk factors.

The economic magnitude of the LEI effect remains substantial after accounting for transaction costs, with net returns of 22 basis points per month and a t-statistic of 2.92 for the baseline value-weighted NYSE-breakpoint strategy. Among large-cap stocks exceeding the 80th NYSE size percentile, the LEI strategy delivers an average return of 36 basis points per month with a t-statistic of 4.01, confirming that the predictive power extends beyond small, illiquid securities. The strategy’s performance proves robust to alternative portfolio construction methodologies, with twenty-one out of twenty-five specification combinations producing alphas with t-

statistics exceeding three.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the LEI trading strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.67. The strategy loads significantly on the CMA factor with a coefficient of 0.27 (t-statistic of 5.39), consistent with the production-based interpretation that leverage expansion intensity captures variation in firms' investment frictions. These results establish LEI as an economically meaningful predictor that ranks in the top 6% of documented anomalies when accounting for implementation costs.

This paper contributes to the production-based asset pricing literature by establishing leverage expansion intensity as a powerful predictor of cross-sectional returns that directly maps to firms' marginal costs and investment frictions. While [Fama and French \(2015\)](#) document the role of investment and profitability in explaining average returns, and [Hou et al. \(2015\)](#) develop the investment CAPM framework, our work demonstrates that the intensive margin of leverage adjustment provides distinct information about firms' production efficiency and risk exposure. Unlike static leverage measures examined in [Bhandari \(1988\)](#) or broad financing variables studied in [Bradshaw et al. \(2006\)](#), LEI captures the dynamic optimization of capital structure that reveals time-varying exposure to productivity shocks.

Our methodological innovation lies in isolating the intensity component of leverage changes, which proves more informative than level measures or simple growth rates examined in prior work such as [Cooper et al. \(2008\)](#) on asset growth or [Lyandres and Palazzo \(2016\)](#) on net debt issuance. The LEI signal achieves a net Sharpe ratio of 0.41, placing it in the 96th percentile among 212 documented anomalies after accounting for transaction costs, demonstrating exceptional out-of-sample performance using the rigorous testing protocol of [Novy-Marx and Velikov \(2023b\)](#). This superior performance stems from LEI's ability to capture firms' marginal productivity of cap-

ital and their exposure to aggregate investment shocks more precisely than existing measures.

The broader implications extend our understanding of how production decisions generate cross-sectional variation in expected returns through rational risk channels. By documenting that leverage expansion intensity predicts returns even among large, liquid stocks and after controlling for known factors, we provide evidence that financial markets efficiently incorporate information about firms' production functions and adjustment costs. These findings support production-based models where cross-sectional differences in expected returns arise from heterogeneous exposure to systematic productivity and investment shocks, offering a unified framework for understanding the joint dynamics of corporate financing, real investment, and asset prices.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key components needed to construct our leverage expansion intensity measure. signal relies on two primary accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's debt maturity profile beyond one year. Invested capital (ICAPT) measures the total funds invested in the company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable provides a comprehensive measure

of the firm’s capital base that supports its operating activities. construct the Leverage Expansion Intensity signal by computing the year-over-year change in long-term debt issuance, scaled by lagged invested capital: $(DLTIS(t) - DLTIS(t-1)) / ICAPT(t-1)$. This ratio captures the acceleration or deceleration in a firm’s long-term borrowing activity relative to its existing capital base. A positive value indicates that the firm is increasing its pace of long-term debt issuance compared to the previous year, while a negative value suggests a reduction in debt issuance intensity. We calculate this measure using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing values for DLTIS in consecutive years and positive values for lagged ICAPT. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LEI signal. Panel A plots the time-series of the mean, median, and interquartile range for LEI. On average, the cross-sectional mean (median) LEI is -0.04 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LEI data. The signal’s interquartile range spans -0.11 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the LEI signal for the CRSP universe. On average, the LEI signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high LEI portfolio and sells the low LEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LEI strategy earns an average return of 0.27% per month with a t-statistic of 3.73. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.25% to 0.32% per month and have t-statistics exceeding 3.39 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 3.99. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-24bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.29. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LEI, as well as average returns and alphas for long/short trading LEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LEI strategy achieves an average return of 36 bps/month with a t-statistic of 4.01. Among these large cap stocks, the alphas for the LEI strat-

egy relative to the five most common factor models range from 28 to 39 bps/month with t-statistics between 3.09 and 4.36.

5 How does LEI perform relative to the zoo?

Figure 2 puts the performance of LEI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LEI strategy falls in the distribution. The LEI strategy’s gross (net) Sharpe ratio of 0.52 (0.41) is greater than 91% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LEI strategy (red line).² Ignoring trading costs, a \$1 invested in the LEI strategy would have yielded \$3.64 which ranks the LEI strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LEI strategy would have yielded \$2.25 which ranks the LEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LEI relative to those. Panel A shows that the LEI strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LEI ranks between the 77 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LEI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LEI or at least to weaken the power LEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LEI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LEI signal in these Fama-MacBeth regressions exceed 1.50, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on LEI is 0.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the LEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LEI strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.78, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LEI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.67.

7 Does LEI add relative to the whole zoo?

Finally, we can ask how much adding LEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LEI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LEI grows to \$1135.26.

8 Conclusion

This study provides compelling evidence that Leverage Expansion Intensity (LEI) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LEI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LEI delivers an annualized gross Sharpe ratio of 0.52, which remains economically mean-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LEI is available.

ingful at 0.41 after accounting for transaction costs. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (25 bps/month gross, 20 bps/month net) underscores its distinct information content beyond conventional risk factors. The persistence of a 19 bps monthly alpha even after controlling for six closely related financing and investment strategies—including changes in financial liabilities, net debt financing, and inventory growth—suggests that LEI captures unique aspects of firm financing behavior not fully reflected in existing measures.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The strategy’s net Sharpe ratio of 0.41 indicates that LEI can be profitably implemented even after realistic transaction costs, making it a viable addition to quantitative investment strategies. The signal’s robustness across multiple risk adjustments suggests it may help investors better understand and exploit inefficiencies related to corporate leverage decisions.

However, several limitations warrant consideration. First, while our analysis controls for major risk factors and related strategies, the possibility remains that LEI proxies for an unidentified systematic risk factor. Second, the implementation of LEI strategies may face capacity constraints in practice, particularly for large institutional investors, which could erode the documented returns. Third, our sample period and geographic focus may limit the generalizability of findings to other markets or time periods.

Future research should explore several promising avenues. Investigating the economic mechanisms underlying LEI’s predictive power could provide deeper insights into why leverage expansion intensity predicts returns. Examining the signal’s performance across different market conditions, particularly during financial crises or periods of monetary policy shifts, would enhance our understanding of its robustness. Additionally, exploring potential interactions between LEI and other corporate

decisions, such as investment policy or payout choices, could reveal important cross-effects. Finally, extending the analysis to international markets would help establish whether the LEI anomaly represents a global phenomenon or is specific to certain market structures or regulatory environments.

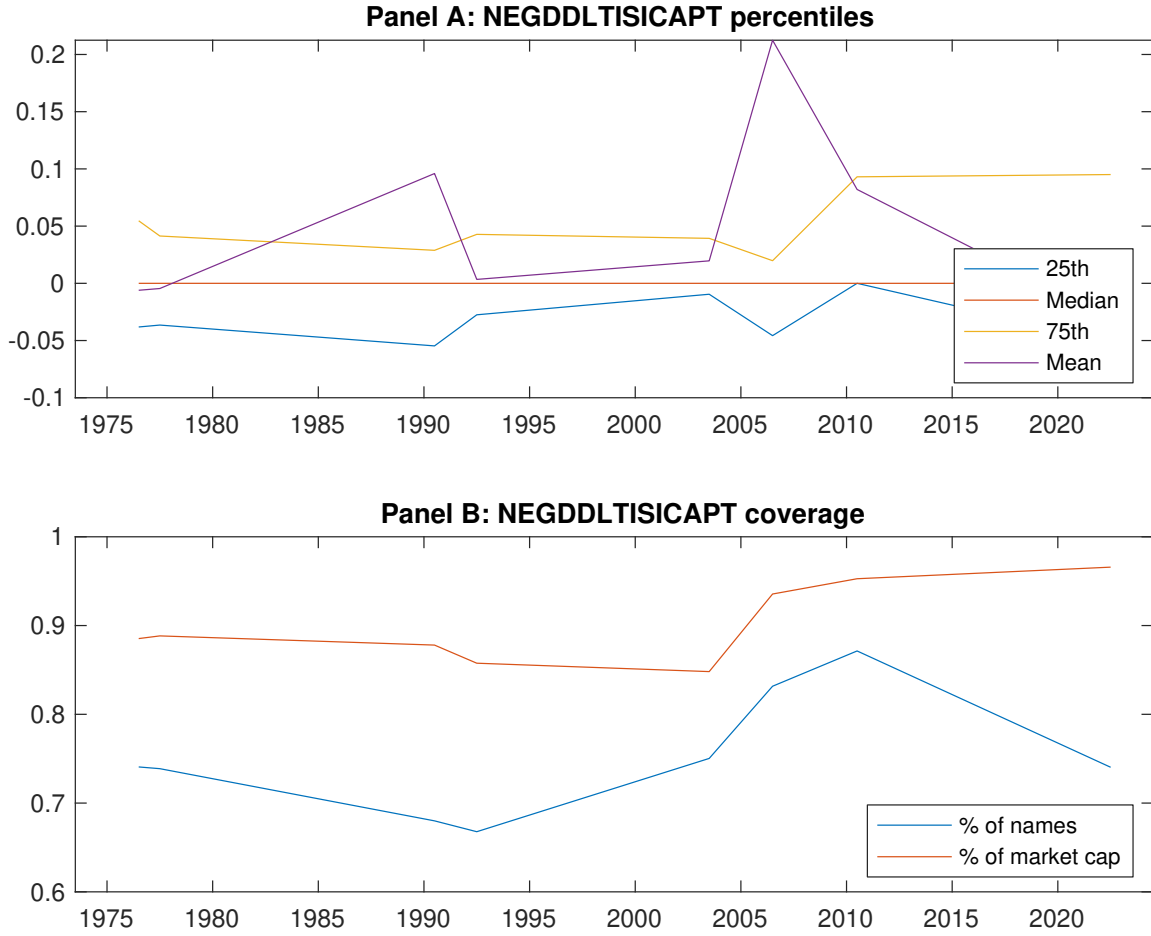


Figure 1: Times series of LEI percentiles and coverage.
This figure plots descriptive statistics for LEI. Panel A shows cross-sectional percentiles of LEI over the sample. Panel B plots the monthly coverage of LEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.62]	0.66 [3.68]	0.64 [3.24]	0.72 [4.11]	0.83 [4.13]	0.27 [3.73]
α_{CAPM}	-0.17 [-3.14]	0.05 [1.05]	-0.03 [-0.43]	0.13 [2.68]	0.15 [2.70]	0.32 [4.33]
α_{FF3}	-0.17 [-3.27]	0.01 [0.15]	0.03 [0.57]	0.10 [2.25]	0.15 [2.78]	0.32 [4.40]
α_{FF4}	-0.14 [-2.65]	0.02 [0.58]	0.07 [1.17]	0.07 [1.56]	0.14 [2.51]	0.28 [3.77]
α_{FF5}	-0.17 [-3.29]	-0.03 [-0.81]	0.09 [1.55]	0.01 [0.13]	0.10 [1.89]	0.28 [3.78]
α_{FF6}	-0.15 [-2.85]	-0.02 [-0.42]	0.11 [1.93]	-0.01 [-0.23]	0.10 [1.79]	0.25 [3.39]
Panel B: Fama and French (2018) 6-factor model loadings for LEI-sorted portfolios						
β_{MKT}	1.08 [86.99]	0.98 [98.38]	0.98 [73.95]	0.96 [94.17]	1.03 [80.69]	-0.04 [-2.49]
β_{SMB}	0.11 [5.94]	-0.12 [-8.01]	-0.02 [-1.07]	-0.03 [-1.91]	0.13 [6.92]	0.02 [0.88]
β_{HML}	0.02 [0.92]	0.15 [7.58]	-0.11 [-4.31]	-0.01 [-0.52]	-0.10 [-4.12]	-0.12 [-3.74]
β_{RMW}	0.10 [4.13]	0.10 [5.30]	-0.05 [-2.05]	0.11 [5.69]	0.04 [1.41]	-0.06 [-1.93]
β_{CMA}	-0.12 [-3.33]	0.02 [0.65]	-0.13 [-3.46]	0.21 [7.16]	0.15 [4.01]	0.27 [5.39]
β_{UMD}	-0.04 [-2.94]	-0.03 [-2.65]	-0.04 [-2.68]	0.02 [2.45]	0.01 [0.56]	0.04 [2.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	526	1062	584	648	
me (\$10 ⁶)	1494	3076	2226	3098	1458	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.73]	0.32 [4.33]	0.32 [4.40]	0.28 [3.77]	0.28 [3.78]	0.25 [3.39]
Quintile	NYSE	EW	0.18 [3.99]	0.21 [4.58]	0.20 [4.29]	0.18 [3.78]	0.19 [3.97]	0.17 [3.63]
Quintile	Name	VW	0.26 [3.71]	0.32 [4.57]	0.33 [4.74]	0.28 [3.97]	0.27 [3.92]	0.24 [3.44]
Quintile	Cap	VW	0.23 [3.48]	0.27 [4.17]	0.28 [4.22]	0.23 [3.43]	0.19 [2.99]	0.16 [2.50]
Decile	NYSE	VW	0.31 [3.38]	0.36 [3.92]	0.35 [3.79]	0.36 [3.81]	0.23 [2.47]	0.25 [2.70]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [2.92]	0.25 [3.42]	0.25 [3.46]	0.23 [3.13]	0.22 [2.97]	0.20 [2.72]
Quintile	NYSE	EW	-0.08 [-1.29]					
Quintile	Name	VW	0.21 [2.86]	0.26 [3.76]	0.27 [3.89]	0.25 [3.50]	0.23 [3.26]	0.20 [2.97]
Quintile	Cap	VW	0.18 [2.69]	0.22 [3.43]	0.23 [3.45]	0.20 [3.04]	0.16 [2.44]	0.14 [2.12]
Decile	NYSE	VW	0.24 [2.57]	0.29 [3.09]	0.28 [2.98]	0.29 [3.02]	0.18 [1.92]	0.18 [1.97]

Table 3: Conditional sort on size and LEI

This table presents results for conditional double sorts on size and LEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LEI and short stocks with low LEI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LEI Quintiles					LEI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.46]	0.91 [3.30]	0.89 [3.30]	0.88 [3.19]	0.80 [2.87]	0.13 [1.48]	0.15 [1.72]	0.14 [1.60]	0.09 [1.03]	0.11 [1.27]	0.08 [0.91]
	(2)	0.71 [2.66]	0.90 [3.62]	0.84 [3.31]	0.87 [3.53]	0.87 [3.44]	0.16 [2.00]	0.20 [2.47]	0.17 [2.20]	0.17 [2.14]	0.13 [1.70]	0.14 [1.72]
	(3)	0.80 [3.17]	0.79 [3.60]	0.82 [3.42]	0.82 [3.66]	0.88 [3.75]	0.09 [1.09]	0.14 [1.85]	0.13 [1.68]	0.09 [1.09]	0.12 [1.45]	0.08 [1.04]
	(4)	0.72 [3.11]	0.81 [3.92]	0.82 [3.67]	0.73 [3.52]	0.92 [4.24]	0.20 [2.63]	0.24 [3.22]	0.23 [3.07]	0.19 [2.46]	0.21 [2.66]	0.18 [2.23]
	(5)	0.47 [2.38]	0.63 [3.50]	0.59 [3.06]	0.60 [3.31]	0.83 [4.33]	0.36 [4.01]	0.38 [4.24]	0.39 [4.36]	0.33 [3.66]	0.32 [3.52]	0.28 [3.09]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LEI Quintiles					LEI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	392	393	393	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	62	60	61	61	
	(3)	76	77	76	77	76	107	110	105	106	108	
	(4)	64	65	64	65	64	229	241	229	234	231	
(5)	59	59	59	59	59	1452	2217	1819	2183	1506		

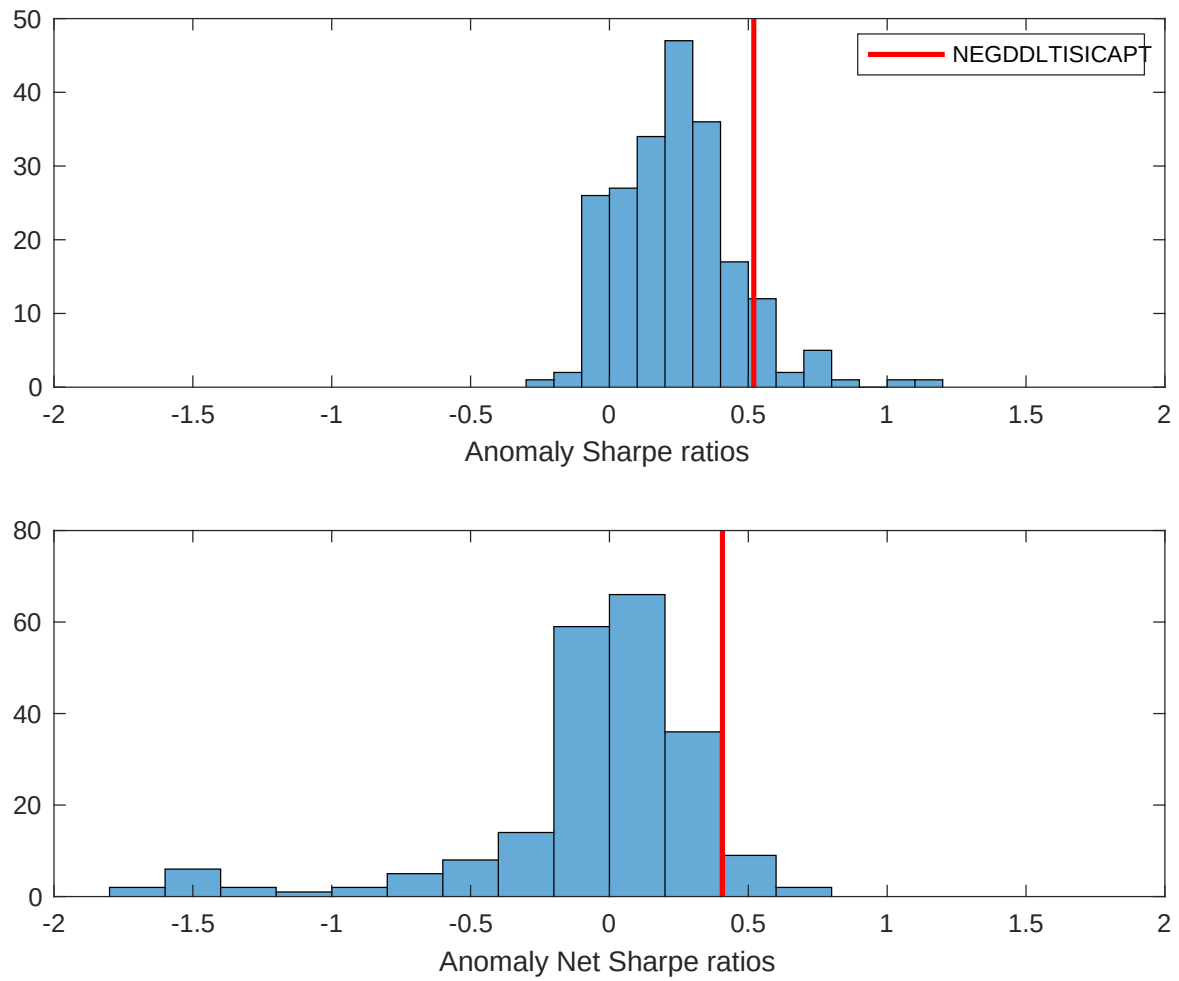


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

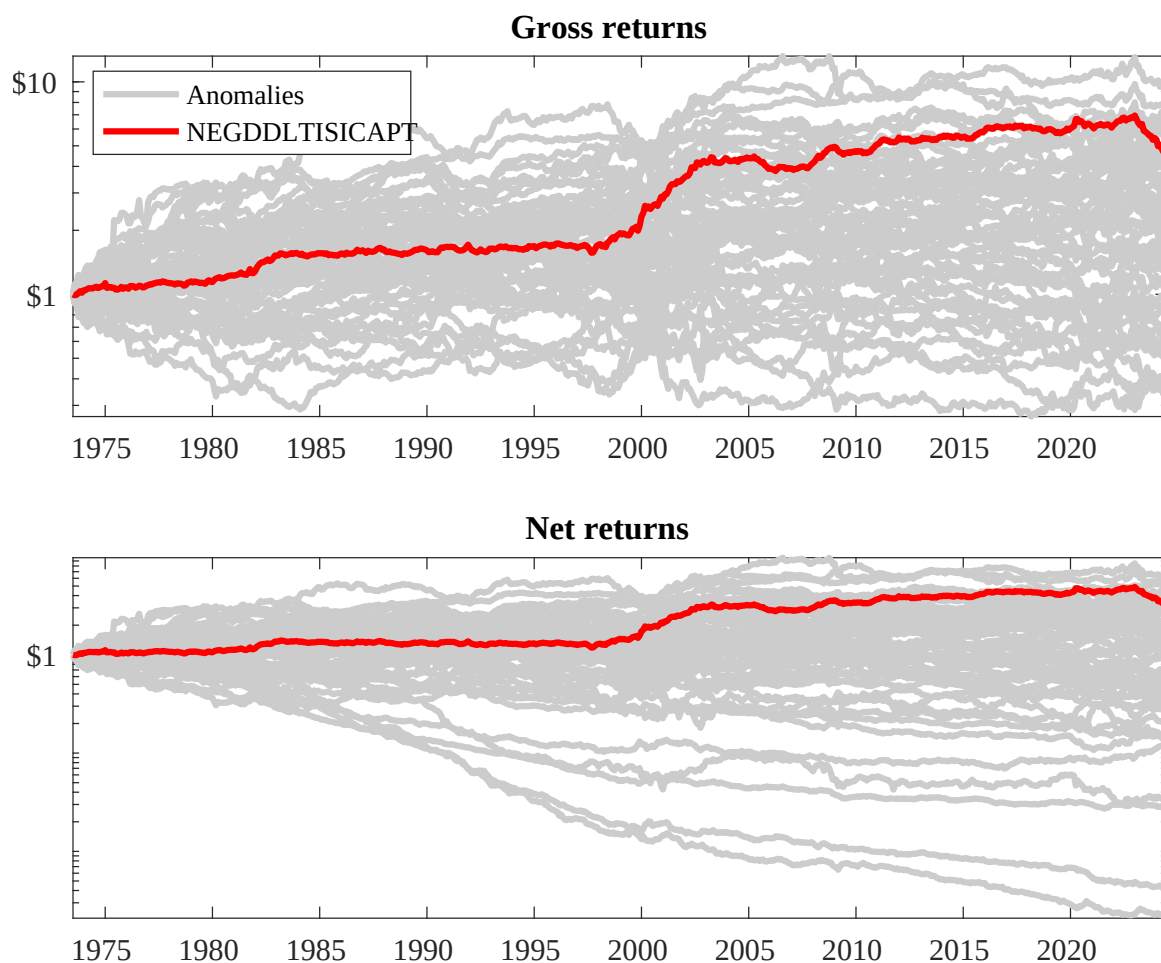
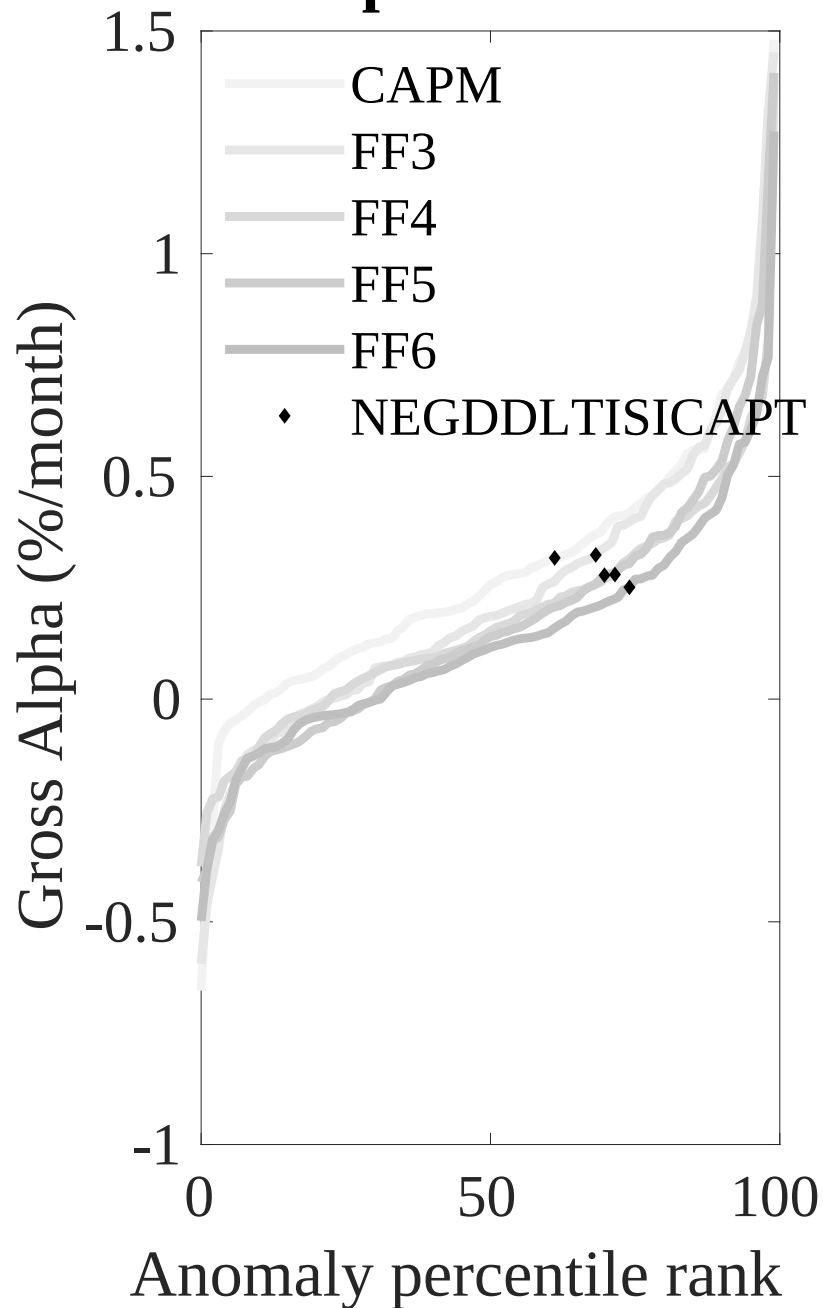


Figure 3: Dollar invested.

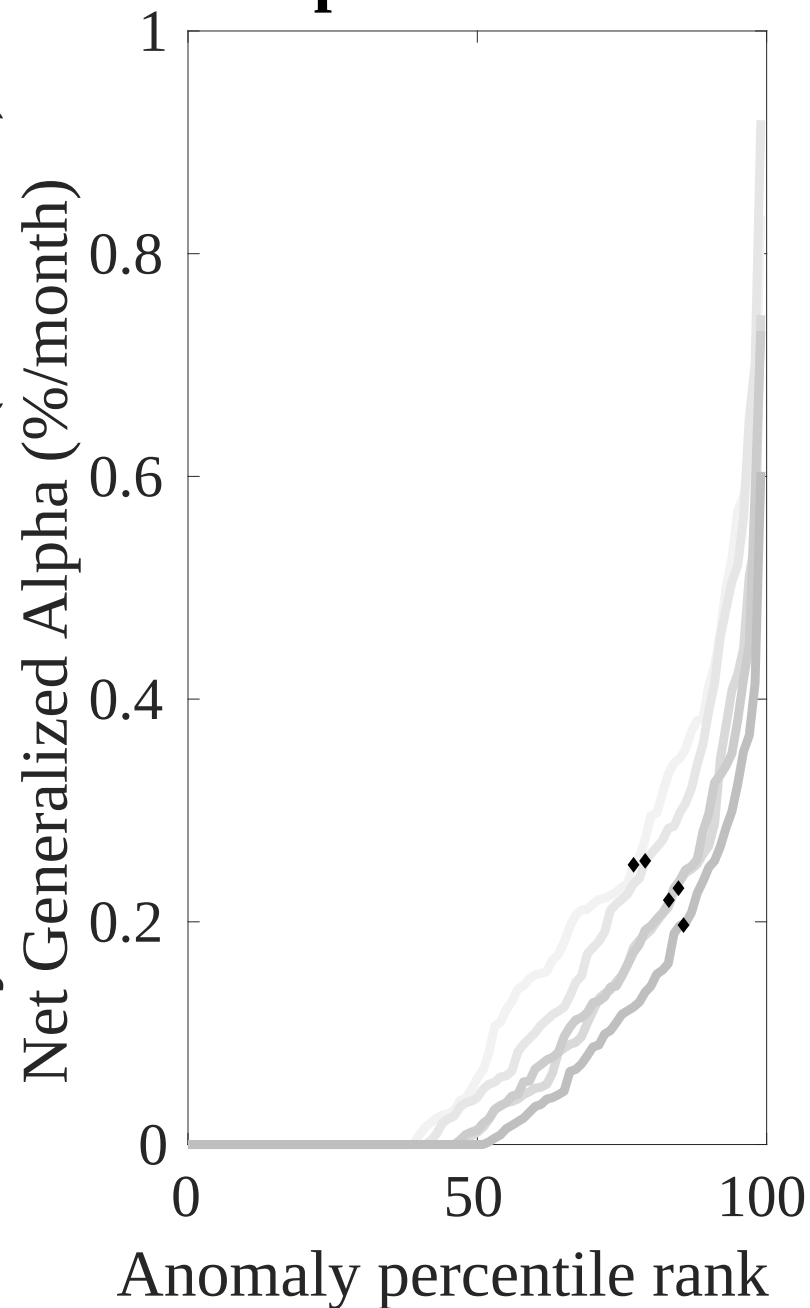
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



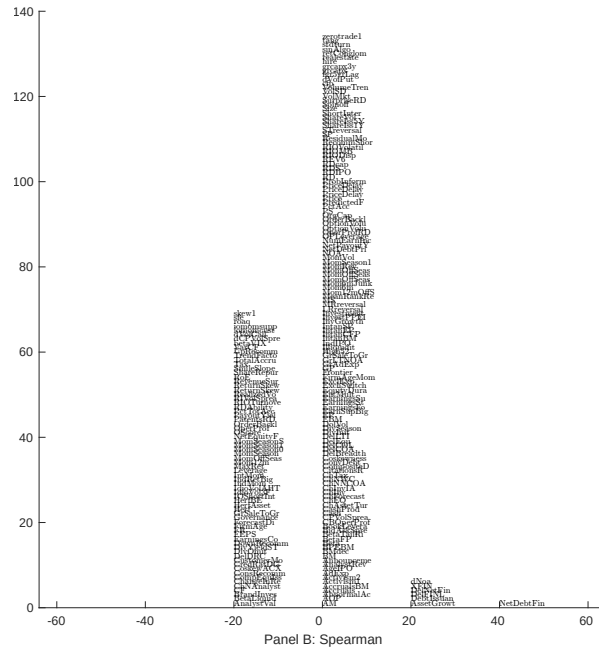
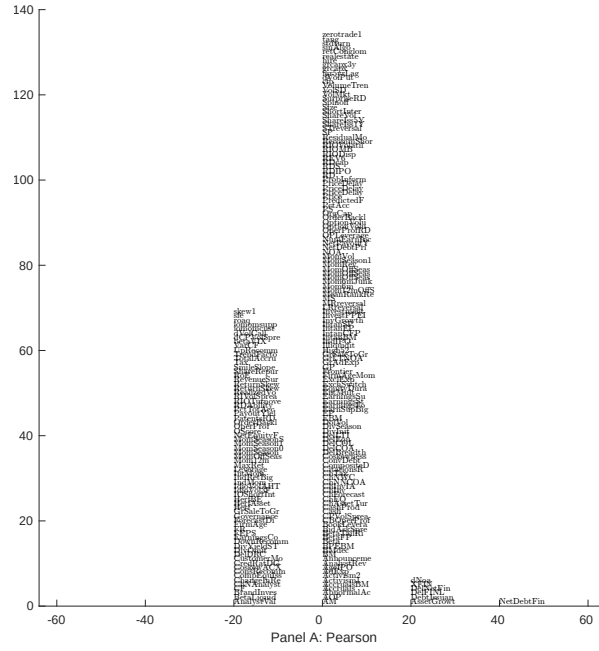


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with LEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

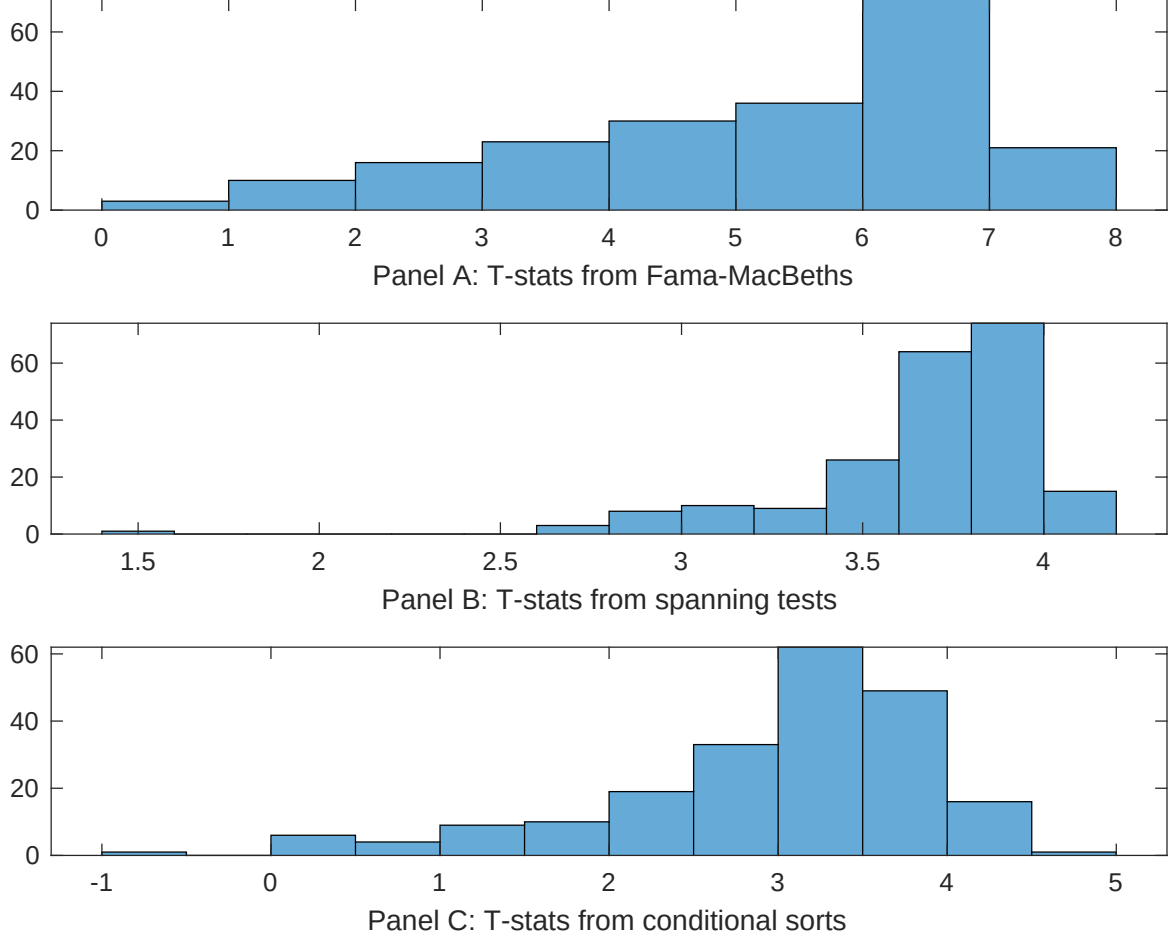


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.37]	0.13 [5.33]	0.14 [5.32]	0.14 [5.74]	0.14 [5.69]	0.13 [5.25]	0.15 [5.72]
LEI	0.89 [1.50]	0.10 [1.58]	0.34 [5.10]	0.16 [2.54]	0.16 [2.35]	0.29 [4.66]	0.11 [0.14]
Anomaly 1	0.17 [9.07]						-0.24 [-0.05]
Anomaly 2		0.20 [8.42]					0.86 [1.29]
Anomaly 3			0.40 [6.86]				0.69 [1.19]
Anomaly 4				0.17 [7.62]			0.13 [4.10]
Anomaly 5					0.19 [6.09]		0.13 [2.36]
Anomaly 6						0.61 [4.03]	-0.52 [-1.50]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LEI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22 [3.00]	0.22 [3.07]	0.24 [3.32]	0.24 [3.26]	0.22 [3.02]	0.20 [2.78]	0.19 [2.67]
Anomaly 1	18.41 [4.47]						2.01 [0.35]
Anomaly 2		21.09 [5.34]					10.96 [2.05]
Anomaly 3			8.03 [2.81]				7.25 [2.48]
Anomaly 4				7.03 [2.12]			-1.94 [-0.55]
Anomaly 5					14.86 [4.07]		11.33 [2.98]
Anomaly 6						17.72 [5.17]	13.32 [3.44]
mkt	-3.54 [-2.11]	-3.92 [-2.36]	-4.11 [-2.43]	-4.06 [-2.40]	-1.92 [-1.10]	-3.63 [-2.18]	-2.17 [-1.26]
smb	0.61 [0.24]	0.63 [0.25]	3.38 [1.31]	2.52 [0.99]	7.27 [2.60]	3.53 [1.40]	6.51 [2.26]
hml	-10.91 [-3.37]	-11.28 [-3.52]	-12.39 [-3.80]	-12.88 [-3.92]	-10.48 [-3.22]	-12.45 [-3.88]	-10.61 [-3.30]
rmw	-7.89 [-2.40]	-8.55 [-2.61]	-5.07 [-1.52]	-6.11 [-1.84]	-15.12 [-3.83]	-4.11 [-1.25]	-11.78 [-2.98]
cma	21.24 [4.18]	21.94 [4.41]	20.26 [3.65]	21.83 [3.89]	17.71 [3.25]	31.93 [6.45]	15.06 [2.37]
umd	2.91 [1.71]	3.14 [1.88]	3.68 [2.16]	4.49 [2.67]	4.40 [2.64]	4.20 [2.54]	2.56 [1.52]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	14	11	10	12	13	17

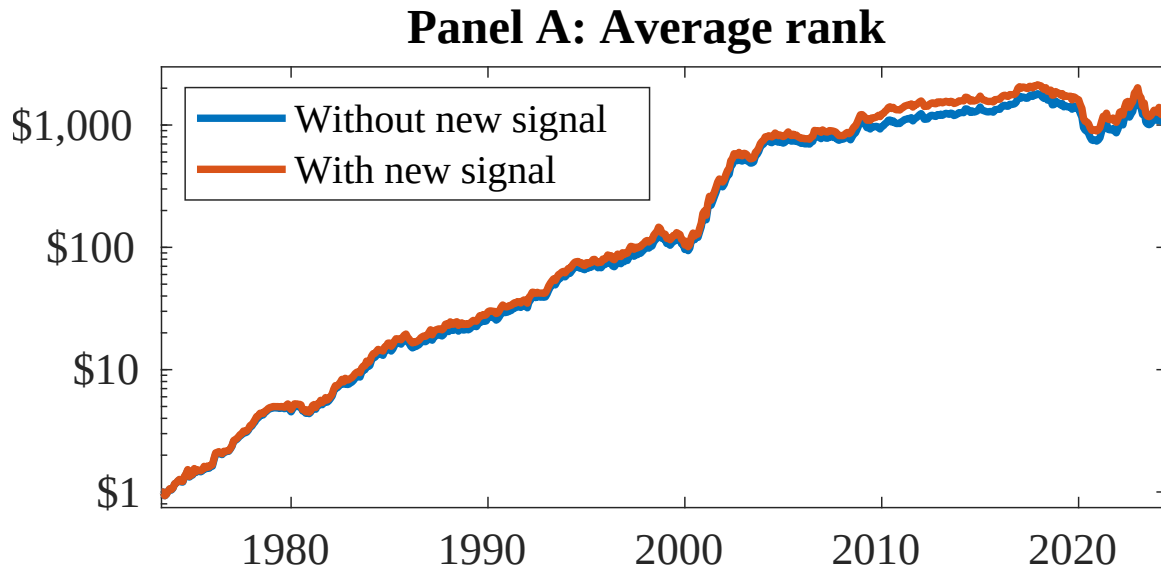


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LEI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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